Docusign Envelope ID: 0F878412-379B-4C38-85C3-93348E32F5B2

PACKAGING AUTOMATION

## | CONNECTIONS

CMC will provide a CIS protocol. This document contains all information related the integration of the CMC line with an external data manager **C**ustomer **I**nformation **S**ystem (CIS)

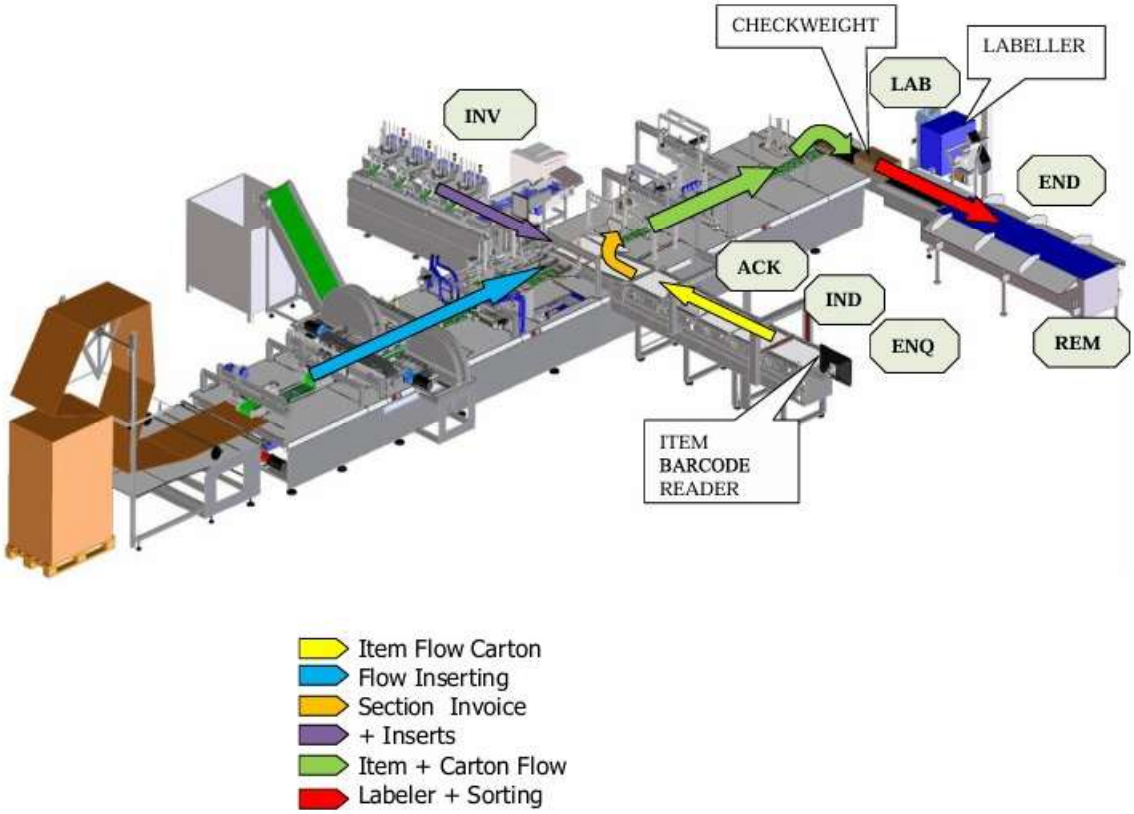
All necessary customer-specific requirements will be defined in the CIS protocol and are included in the total net price.

Furthermore, the package dimensions (L, W, H) and the item weights will be reported back to Conrad's WMS / SAP system.

A direct communication between SAP / WMS system and the shipping labeler of the CartonWrap XL machine is expected to be possible, but must be checked in detail during project phase.
Alternative a middleware software can be implemented via an external provider.
Any additional costs are not part of the offer and must be discussed and ordered separately.

**PRODUCTION FLOW Carton Wrap**

As for evaluation, the following diagram is related to CW1000



Docusign Envelope ID: 0F878412-379B-4C38-85C3-93348E32F5B2



The following documentation is related the EXTERNAL MODE
When the operator 'read' the barcode on Item by hand scanner or optionally by camera On Top, CMC will send a message **'ENQ'** (Enquire) to 'CIS' containing the barcode just scanned. 'CIS' will reply to CMC with a MESSAGE **'enq'** containing all product information related this barcode (flag ValidCode 'YES/NO', description etc.. see below).

If Flag ValidCode is YES (the item is considered valid by CIS), the incoming belt will start the movement (yellow arrow flow) and the item is inducted in the machine. CMC will inform CIS about the item just started with the message **'IND'** (inducted).

CMC will now measure the maximum 3 sizes (L, H, W) of item just inducted using the 3D dimensional bars. Just after the 3D bars, CMC will send the message **'ACK'** to inform CIS if item is 'accepted' or not (may rejected because out of range).

If size of item is out of range (some sizes exceed the maximum allowed for this carton width), the CMC will divert automatically this item without stopping line, on the gate just after the measurement bar.

If the item has been accepted, the CMC will proceed with Creation of Box on Forming section (blue arrow flow), will insert the item inside the box just composed (orange arrow flow), will close the box (green arrow flow) and will delivery on output belt.

Optionally (if installed), at the correct time CMC will request to print the invoice with the MESSAGE **'INV'**; in this case, 'CIS' need to send the data formatted directly to the laser printer (see above).

When the box is at the end of the line, just in front of thermal labeler, the CMC send **MESSAGE 'LAB'** to 'CIS' within initial bar code, ReferenceID, box status (good or bad) and all others data related this box (sizes, weight etc.) 'CIS' will send a formatted label data directly to shipping label printer (see below).

After the label is printed and applied, a bar code reader will verify the correct matching; CMC will send **MESSAGE 'END'** within final box status and all information about box (good or rejected, sizes etc.).

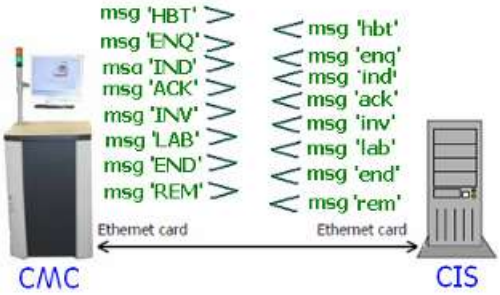
Note: If the Item is "accepted" by the 3D bars, and has been removed manually from the machine, CMC will send the **MESSAGE 'REM'** (removed) to indicate that the item is not passed through the end of the machine.

CIS can check the current status of CMC at any time using the message **'STS'** (status). CMC will return and answer with the current status.

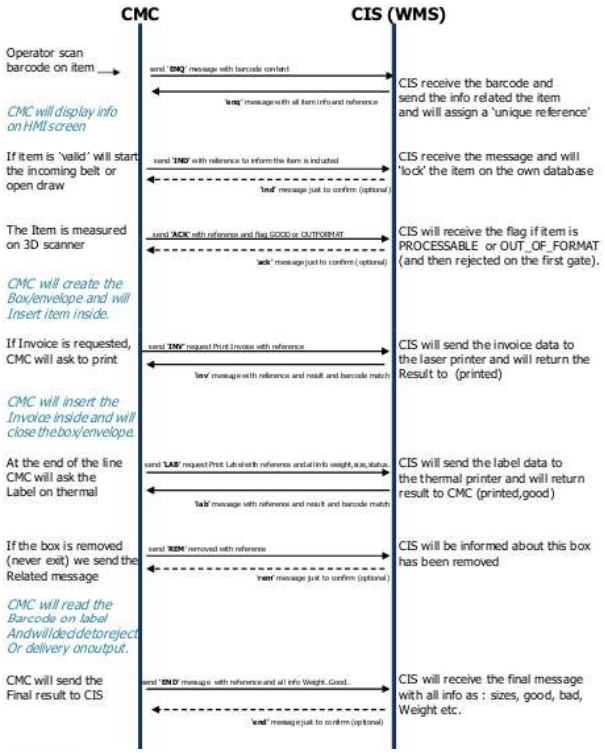
At any time, as debug or troubleshooting, there is an 'heartbeat' message to check the connection between systems. This message can be generated and sent 'on demand' either from CIS to CMC or from CMC to CIS.

Docusign Envelope ID: 0F878412-379B-4C38-85C3-93348E32F5B2

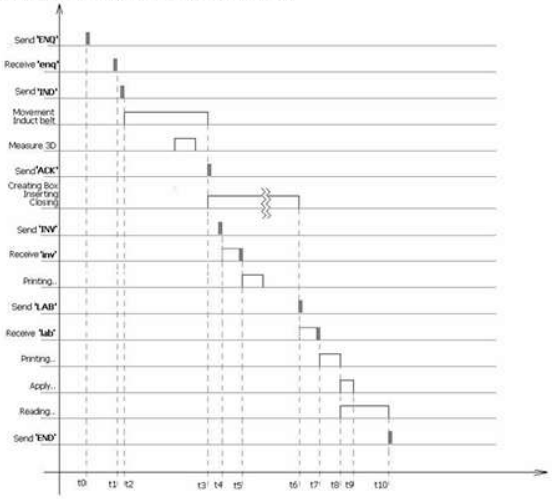
PACKAGING AUTOMATION




**MESSAGE FLOW Carton Wrap**




**MESSAGE TIMING Carton Wrap**




**t0** Time when the operator 'scan' the item
**t1** Time when CMC receive the info related the item
**t2** Time when the item start the induction (if product is valid) **t3** Time when item has been measured and size are checked **t4** Time when CMC require to print the Invoice (optional)
**t5** Time when the customer have sent the invoice data to the printer
**t6** Time when the box exit from the machine and CMC require the label
**t7** Time when the customer have sent the label data to thermal printer
**t8** Time when the printer have received data and printed the label
**t9** Time when the label has been applied
**t10** Time when the barcode reader have verified the label correct